Jiarong Lin

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Github | Google-Scholar | Ziv-Lin-LJR

Associate Professor , Institute of Unmanned Systems , **Beihang University (BUAA)**

Work Experiences

- **Beihang University (BUAA)** *May 2025 - Now*
  Associate Professor Beijing, China
- **DiDi autonomous driving** *May 2024 - May 2025*
  Senior Research and Development Engineer Guangzhou, China
- **Da Jiang Innovations (DJI)** *July 2015 - August 2018*
  Computer vision engineer Shenzhen, China

Education

- **The University of Hong Kong (HKU)** *January 2019 - December 2023*
  Ph.D. in Robotics Hong Kong SAR, China
  - **Specialization**: Robotics; LiDAR SLAM; Sensor fusion.
  - **Laboratory**: Mechatronics and Robotic Systems (MaRS) Laboratory.
  - **Supervisor**: Fu Zhang
- **The Hong Kong University of Science and Technology (HKUST)** *August 2018 - January 2019*
  Ph.D. student Hong Kong SAR, China
  - **Specialization**: Robotics; UAV control; Deep reinforcement learning.
  - **Supervisor**: Fu Zhang
- **University of Electronic Science and Technology of China (UESTC)** *September 2011 - June 2015*
  Bachelor of Science Chengdu, China
  - **Specialization**: Optical Information Science and Technology

Academic Service

- **Associate Editor:**
  - *IEEE International Conference on Robotics and Automation (**ICRA**),*
  - *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*
- **Journal/Conference Reviewer:**
  ***T-RO**, **IJRR**, **RA-L**, **T-IV**, **ICRA**, **IROS**, **RSS**, ...*

Honors and Awards

- **Best Paper Award** 2025
  King-Sun Fu Memorial IEEE Transactions on Robotics (T-RO) Best Paper Award for "FAST-LIVO2: Fast, Direct LiDAR-Inertial-Visual Odometry."
- **Best Paper Award Finalist** 2026
  IEEE/RSJ IROS Best Paper Award on Safety, Security, and Rescue Robotics. "AUG: Perception-Aware UAV Graph-Based Planning for GNSS-Denied Navigation in Feature-Sparse Environments."
- **Best Thesis Award** 2022–2023
  Department of Mechanical Engineering, The University of Hong Kong. Awarded to one recipient annually.
- **World's Top 2% Scientists** 2025, 2026
  Selected for two consecutive years in the ***single-year impact category*** of the world's top 2% scientists list compiled by ***Stanford University and Elsevier***.

Publications (5,630 Citations & H-Index of 19) Google-Scholar

Journal (First author of 1×*T-PAMI*, 2×*T-RO*, 1×*RA-L*)

[1] **J. Lin**, and F. Zhang, "R^{3}LIVE++: A Robust, Real-time, Radiance reconstruction package with a tightly-coupled LiDAR-Inertial-Visual state Estimator," *IEEE Transactions on Pattern Analysis and Machine Intelligence (**T-PAMI**, IF=20.8)*, DOI: 10.1109/TPAMI.2024.3456473

[2] **J. Lin**, C Yuan, Y. Cai, H. Li, Y. Zou, X. Hong and F. Zhang, "ImMesh: An Immediate LiDAR Localization and Meshing Framework," *IEEE Transactions on Robotics (**T-RO**, IF=9.4)*, DOI: 10.1109/TRO.2023.3321227.

[3] C Yuan, **J. Lin (co-first author)** , Z. Liu, X. Hong and F. Zhang, "BTC: A Binary and Triangle Combined Descriptor for 3D Place Recognition," *IEEE Transactions on Robotics (**T-RO**, IF=9.4)*, DOI: 10.1109/TRO.2024.3353076.

[4] **J. Lin**, C. Zheng, W. Xu, and F. Zhang, "R^{2}LIVE: A Robust, Real-Time, LiDAR-Inertial-Visual Tightly-Coupled State Estimator and Mapping," *IEEE Robotics and Automation Letters (**RA-L**, IF=4.6)*, DOI: 10.1109/LRA.2021.3095515

[5] Y Ren, F Zhu, G Lu, Y Cai, L Yin, F Kong, **J Lin**, N Chen, F Zhang. "Safety-assured high-speed navigation for MAVs". *Science Robotics (**Sci-Ro**, IF=26.1)*, DOI: 10.1126/scirobotics.ado6187

[6] H. Li, Y. Zhou, N. Chen, **J. Lin**, et al. "MARS-LVIG dataset: A multi-sensor aerial robots SLAM dataset for LiDAR-visual-inertial-GNSS fusion". *The International Journal of Robotics Research (**IJRR**, IF=9.2)*, DOI: 10.1177/02783649241227968

[7] W. Xu*, Y. Cai*, D. He, **J. Lin**, and F. Zhang, "FAST-LIO2: Fast Direct LiDAR-Inertial Odometry," *IEEE Transactions on Robotics (**T-RO**, IF=9.4)*, DOI: 10.1109/TRO.2022.3141876

[8] C Zheng, W Xu, Z Zou, T Hua, C Yuan, D He, B Zhou, Z Liu, **J Lin**, F Zhu, et al. "Fast-livo2: Fast, direct lidar-inertial-visual odometry," *IEEE Transactions on Robotics (**T-RO**, IF=9.4)*, DOI: 10.1109/TRO.2024.3502198

[9] Y Cai*, F Kong*, Y Ren, F Zhu, **J Lin**, and F Zhang, "Occupancy Grid Mapping without Ray-Casting for High-resolution LiDAR Sensors", *IEEE Transactions on Robotics (**T-RO**, IF=9.4)*, DOI: 10.1109/TRO.2023.3323936

[10] F. Kong, X. Liu, B. Tang, **J. Lin**, Y. Ren, Y. Cai, F. Zhu, N. Chen, and F. Zhang, "MARSIM: A light-weight point-realistic simulator for LiDAR-based UAVs," *IEEE Robotics and Automation Letters (**RA-L**, IF=4.6)*, DOI: 10.1109/LRA.2023.3264163

Conference (First author of 3×*ICRA*, 2×*IROS*)

[9] **J. Lin**, and F. Zhang, "R^{3}LIVE: A Robust, Real-time, RGB-colored, LiDAR-Inertial-Visual tightly-coupled state Estimation and mapping package," *IEEE International Conference on Robotics and Automation (**ICRA**)*.

[10] **J. Lin**, X. Liu, and F. Zhang, "A decentralized framework for simultaneous calibration, localization and mapping with multiple LiDARs," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*.

[11] **J. Lin**, and F. Zhang, "Loam_livox: A fast, robust, high-precision LiDAR odometry and mapping package for LiDARs of small FoV," *IEEE International Conference on Robotics and Automation (**ICRA**)*.

[12] **J. Lin**, L. Wang, F. Gao, S. Shen, and F. Zhang, "Flying through a narrow gap using neural network: an end-to-end planning and control approach," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*.

[13] C. Yuan, **J. Lin (co-first author)**, Z. Zou, X. Hong, and F. Zhang, "STD: Stable Triangle Descriptor for 3D place recognition," *IEEE International Conference on Robotics and Automation (**ICRA**)*.

[14] C Lin, **J Lin (Corresponding Author)**, Z Sui, XZ Qu, R Wang, K Sheng, B Zhang "An End-to-End Learning-Based Multi-Sensor Fusion for Autonomous Vehicle Localization,"*IEEE International Conference on Robotics and Automation (**ICRA**)*.

[15] R Li, X Liu, H Li, Z Liu, **J Lin**, Y Cai, F Zhang "LVBA: LiDAR-Visual Bundle Adjustment for RGB Point Cloud Mapping,"*IEEE International Conference on Robotics and Automation (**ICRA**)*.

[16] X. Chen, B. Zhou, **J. Lin**, Y. Zhang, F. Zhang, S. Shen, "Fast 3D Sparse Topological Skeleton Graph Generation for Mobile Robot Global Planning," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*.

[17] W. Gao, **J. Lin**, F. Zhang, and S. Shen, "A Screen-Based Method for Automated Camera Intrinsic Calibration on Production Lines" *IEEE International Conference on Automation Science and Engineering (**CASE**)*.

Open-sourced Projects (Total stars gathered: ≈18.2 k★) Github

- **R^{3}LIVE: A Robust, Real-time, RGB-colored, LiDAR-Inertial-Visual tightly-coupled state Estimation and mapping package**
  Tag: *Multi-sensor Fusion; LiDAR SLAM; 3D-Reconstruction;*
  Contributions: $\geq$ **98%**
  https://github.com/hku-mars/r3live (★ 2.5k)

- **FAST_LIO: A computationally efficient and robust LiDAR-inertial odometry (LIO) package**
  Tag: *Multi-sensor Fusion; LiDAR SLAM;*
  Contributions: $\approx$ **15%**
  https://github.com/hku-mars/FAST_LIO (★ 5293)

- **FAST-LIVO2: Fast, Direct LiDAR-Inertial-Visual Odometry**
  Tag: *Multi-sensor Fusion; LiDAR SLAM;*
  Contributions: $<$ **10%**
  https://github.com/hku-mars/fast-livo2 (★ 4.8k)

- **LOAM-livox: A robust LiDAR Odometry and Mapping (LOAM) package for Livox-LiDAR**
  Tag: *LiDAR SLAM;*
  Contributions: $\geq$ **95%**
  https://github.com/hku-mars/loam_livox (★ 1.6k)

- **R^{2}LIVE: A Robust, Real-time, LiDAR-Inertial-Visual tightly-coupled state Estimator and mapping**
  Tag: *Multi-sensor Fusion; LiDAR SLAM;*
  Contributions: $\geq$ **98%**
  https://github.com/hku-mars/r2live (★ 783)

- **ImMesh: An Immediate LiDAR Localization and Meshing Framework**
  Tag: *LiDAR SLAM; 3D-Reconstruction;*
  Contributions: $\geq$ **45%**
  https://github.com/hku-mars/ImMesh (★ 746)

- **STD: A Stable Triangle Descriptor for 3D place recognition**
  Tag: *LiDAR SLAM; Place Recognition; Loop Detection and Loop Closure*
  Contributions: $\geq$ **45%**
  https://github.com/hku-mars/std (★ 747)

- **Other Works (see my GitHub page: https://github.com/ziv-lin)**
  https://github.com/hku-mars/decentralized_loam (★ 222)
  https://github.com/hku-mars/btc_descriptor (★ 373)
  https://github.com/ziv-lin/rxlive_handheld (★ 149)
  Others ... (★ 974)

Teaching

- Instructor, Autonomous Navigation and Cooperative Localization for UAVs, Beihang University. Autumn, 2026
- Teaching Assistant, Electrical and Electronic Engineering, The University of Hong Kong. Autumn, 2019